A recurrence for four-row arrays increasing in four directions

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Abstract

Let a_n count the fillings of a $4 \times n$ rectangle by $1, \dots, 4n$ that increase to the right, downwards, down-right, and down-left. We prove the second-order polynomial recurrence conjectured for OEIS A181198. A cut immediately before the second row becomes full gives an explicit positive double sum. A polynomial antiderivative converts that sum to an integral; two rational integration-by-parts certificates reduce it to a single polynomial moment and prove a first-order inhomogeneous recurrence. Eliminating its rational source gives the conjectured recurrence. All certificates, boundary terms, and exceptional initial indices are included. Exact symbolic verification and independent direct counting accompany the proof; neither finite numerical agreement nor the conjectured recurrence is used to establish the combinatorial formula.

1 Statement and context

A filling is admissible when every adjacent step in one of the four directions just listed increases its entry, whenever the step remains inside the rectangle. Write a_n for the number of admissible fillings. The down-right condition follows from rightward and downward increase but is retained in the definition.

Theorem 1. *For every integer $n \geq 1$,*

$$\begin{aligned} &3(n+1)(2n+3)(3n+4)(3n+5)(7n^2-1)(n+2)^3 a_{n+2} \\ &\quad - 8(n+1)(2n+1)(4n+3)(4n+5) \\ &\qquad \times (364n^5 + 84n^4 - 1025n^3 - 534n^2 + 157n + 54) a_{n+1} \\ &\quad - 64(2n-1)^2(2n+1)(4n-1)(4n+1)(4n+3)(4n+5) \\ &\qquad \times (7n^2 + 14n + 6) a_n = 0. \end{aligned} \tag{1}$$

The initial values are $a_1 = a_2 = 1$ and $a_3 = 8$.

We prove the following more convenient statement first. For $n \geq 2$, put

$$c_n = \frac{4n-1}{n-1} \frac{(n!)^4}{(4n)!} a_n. \tag{2}$$

Then

$$3(3n+1)(3n+2)c_{n+1} + (n+1)^2 c_n = \frac{3(7n^2-1)}{4n(n-1)(2n-1)^2(2n+1)^2}. \tag{3}$$

The sequence was introduced by Hardin in 2010. OEIS credits the conjectured recurrence to Koutschan in February 2023 [1]. Kauers and Koutschan discuss its profile-counting model in Section 6.4 and conjecture a hypergeometric-sum expression in Conjecture 18 [2]. Sun identifies these arrays with truncated shifted strips and proves results for three rows and, in a different parameter direction, fixed widths four and five [3]. Those results do not give the present four-row recurrence. The primary sources consulted on 28 October 2026 still label the target conjectural. A bounded literature audit located no exact resolution; this does not establish priority. The proof below is self-contained and does not assume any claimed open status.

2 Profiles and a strict-partition product

After the smallest k labels have been placed, let x_i be the occupied prefix length in row i. For a nonempty lower row, its rightmost occupied cell requires the cell above in column $\min(n, x_i + 1)$ to be occupied. Therefore a profile is an ideal precisely when

$$0 \le x_i \le n, \qquad x_{i-1} \ge \min(n, x_i + 1) \quad (i > 1,\ x_i > 0). \tag{4}$$

These inequalities imply every vertical and down-left predecessor condition, while row-prefix closure gives the horizontal conditions. Thus positive lengths are strictly decreasing except for repeated full lengths n; zero lengths may repeat. Admissible fillings correspond bijectively to paths from $(0, 0, 0, 0)$ to (n, n, n, n) that add one cell and satisfy (4) at every step.

Lemma 2. *For a strict partition $\lambda = (\lambda_1 > \cdots > \lambda_\ell > 0)$ of m, the number of paths from the empty partition to λ, adding one cell at a time while keeping the positive parts strict, is*

$$g(\lambda) = \frac{m!}{\prod_i \lambda_i!} \prod_{i<j} \frac{\lambda_i - \lambda_j}{\lambda_i + \lambda_j}. \tag{5}$$

Zero parts are omitted.

Proof. We prove the formula by induction on m. The empty product is one. The ratio of the proposed value at $\lambda - e_i$ to that at λ is A_i/m, where

$$A_i = \lambda_i \prod_{j \ne i} \frac{\lambda_i - \lambda_j - 1}{\lambda_i - \lambda_j} \frac{\lambda_i + \lambda_j}{\lambda_i + \lambda_j - 1}.$$

An illegal decrement producing a repeated positive part has $A_i = 0$. A final part equal to one can become zero and be omitted, with the same ratio. It remains to prove $\sum_i A_i = m$.

Consider the rational function

$$F(z) = \prod_j \frac{(z - \lambda_j - 1)(z + \lambda_j)}{(z - \lambda_j)(z + \lambda_j - 1)}.$$

It satisfies $F(1 - z) = F(z)$ and has possible simple poles at λ_i and $1 - \lambda_i$. The residues of $(z - \frac{1}{2})F(z)$ at each such pair are both $-A_i$. On the other hand, $F(z) = 1 - 2mz^{-2} + O(z^{-3})$. Comparing the coefficient of z^{-1} in the partial-fraction expansion of $(z - \frac{1}{2})F(z)$ gives $-2\sum_i A_i = -2m$. Hence the proposed product obeys exactly the recursion obtained by deleting the last cell, which completes the induction. □

3 A positive double sum

Cut each growth path immediately before the last cell of row two is inserted. For $n \ge 2$ the profile at this unique instant is

$$(n, n - 1, b, c), \qquad 0 \le c < b \le n - 2,$$

or $(n, n - 1, 0, 0)$. Until this instant there is no repeated positive part: two full rows would mean that row two had already finished. Conversely every strict-partition path to this profile is a legal prefix. Its number is $g(n, n - 1, b, c)$, with zero parts omitted.

After the next, specified cell is inserted, only the last two rows remain. For $b > c$, delete the forced final step from $(n, n - 1)$ to (n, n). We must count paths from (b, c) to $(n, n - 1)$ staying

strictly above the diagonal. Reflection at the first collision subtracts the paths starting at (c, b) from all paths starting at (b, c). Thus their number is

$$\begin{aligned} N(n,b,c) &= (2n-b-c-1)!\left\{\frac{1}{(n-b)!(n-c-1)!} - \frac{1}{(n-c)!(n-b-1)!}\right\} \\ &= \frac{(2n-b-c-1)!(b-c)}{(n-b)!(n-c)!}. \end{aligned} \tag{6}$$

For $b = c = 0$, the first step is forced, and the same reflection argument gives $\mathrm{Cat}_{n-1} = \binom{2n-2}{n-1}/n$. Formula (5) gives this same value for the prefix $(n, n-1)$. The cut is reversible and unique, proving

$$a_n = \mathrm{Cat}_{n-1}^2 + \sum_{0 \le c < b \le n-2} g(n, n-1, b, c) N(n, b, c). \tag{7}$$

For $s = b + c > 0$, the summand simplifies to

$$\frac{(2n+s-1)!(2n-s-1)!(b-c)^2 w_b w_c}{(2n-1)n!(n-1)!\,s}, \qquad w_b = \frac{1}{b!(n-b-2)!(n+b)(n+b-1)}. \tag{8}$$

Every factorial has a nonnegative argument on its stated range.

This also gives the initial values without using any recurrence. The case $n = 1$ is a single increasing column. At $n = 2$ the sum is empty. At $n = 3$ the only pair is $(b, c) = (1, 0)$, for which $g(3, 2, 1) = 2$ and $N(3, 1, 0) = 2$. Consequently $a_3 = 2^2 + 2 \cdot 2 = 8$.

4 From the double sum to a polynomial integral

Define

$$W_n(z) = \sum_{b=0}^{n-2} w_b z^b = \frac{1}{(n-2)!} \int_0^1 x^{n-2}(1-x)(1+zx)^{n-2}\,\mathrm{d}x. \tag{9}$$

Write $W = W_n$, $\theta = z\,\mathrm{d}/\mathrm{d}z$, and $L = (n-2)(n-1)$. Expanding the product gives

$$\sum_{b>c} (b-c)^2 w_b w_c z^{b+c} = W\theta^2 W - (\theta W)^2.$$

The ratio of consecutive coefficients in (9) proves

$$z(1+z)W'' + (n+1+2z)W' - LW = 0. \tag{10}$$

Set

$$F_n(z) = \frac{(n-2Lz)W^2 + z(2n+1+2z)WW' + 2z^2(1+z)(W')^2}{2n-1}. \tag{11}$$

Differentiating and using (10) gives

$$F_n' = WW' + z\{WW'' - (W')^2\} = \frac{W\theta^2 W - (\theta W)^2}{z}.$$

Both sides are polynomials, including at $z = 0$. Hence the divided convolution in (8) is $F_n(z) - F_n(0)$.

For positive integers u, v, repeated integration by parts gives

$$\mathrm{B}(u, v) = \int_0^1 t^{u-1}(1-t)^{v-1}\,\mathrm{d}t = \frac{(u-1)!(v-1)!}{(u+v-1)!}.$$

Using $u = 2n + s$, $v = 2n - s$ in (8) therefore gives

$$a_n = \mathrm{Cat}_{n-1}^2 + K_n \int_0^1 t^{2n-1}(1-t)^{2n-1}\left[F_n\left(\frac{t}{1-t}\right) - F_n(0)\right]\,\mathrm{d}t, \tag{12}$$

where $K_n = (4n-1)!/[(2n-1)n!(n-1)!]$. This is integration of a finite sum; its exponents are integrable at both endpoints. Since $W_n(0) = 1/n!$,

$$K_n F_n(0)\,\mathrm{B}(2n,2n) = K_n \frac{n}{(2n-1)(n!)^2}\,\mathrm{B}(2n,2n) = \mathrm{Cat}_{n-1}^2\,.$$

The separate Catalan contribution cancels exactly.

Put

$$\begin{aligned} Y_n(t) &= (1-t)^{n-2} W_n\left(\frac{t}{1-t}\right) \\ &= \frac{1}{(n-2)!}\int_0^1 x(1-x)^{n-2}(1-tx)^{n-2}\,\mathrm{d}x. \end{aligned} \tag{13}$$

This is a polynomial, so the displayed change of variables introduces no endpoint singularity. Substitution in (12) now yields

$$a_n = \frac{K_n}{2n-1}\int_0^1 w_n(t)\mathcal{H}_n(t)\,\mathrm{d}t, \qquad w_n(t) = t^{2n-1}(1-t)^3, \tag{14}$$

where, writing $Y = Y_n$,

$$\mathcal{H}_n = [n+3(n-2)t]Y^2 + t[2n+1+(2n-7)t]YY' + 2t^2(1-t)(Y')^2. \tag{15}$$

All constants in this transformation are retained; in particular, both factors $2n-1$ in (14) and in K_n are needed.

5 Two elementary polynomial identities

For the rest of the proof $n \geq 2$. Define

$$P = t(1-t^2), \qquad R = n + 2(n-2)t + (n-2)t^2, \qquad C = \frac{1}{(n-1)!}.$$

Lemma 3. *The polynomials in (13) satisfy*

$$PY_n' + RY_n = C, \tag{16}$$

$$2t(2n-1)Y_{n+1} + (1-t)^2 Y_n = \frac{1}{n!}. \tag{17}$$

Proof. Put $m = n-2$, $q_t(x) = x(1-t+tx)$, and $J_m(t) = \int_0^1 q_t(x)^m\,\mathrm{d}x$. From (9), or the substitution $x \mapsto 1-x$ in (13),

$$2tm!Y_{m+2} = (1+t)J_m - \frac{1}{m+1}. \tag{18}$$

Indeed integrate $q_t' q_t^m$, using $q_t' = 1-t+2tx$. Since $(q_t')^2 = (1-t)^2 + 4tq_t$, integrating the derivative of $q_t' q_t^{m+1}$ gives

$$2t(2m+3)J_{m+1} + (m+1)(1-t)^2 J_m = 1+t.$$

Substitution of (18) proves (17).

To obtain the differential identity, set $I_m(z) = \int_0^1 [x(1+zx)]^m\,\mathrm{d}x$. The substitution $u = zx$ and differentiation of its upper limit give $zI_m' + (m+1)I_m = (1+z)^m$. As $J_m(t) = (1-t)^m I_m(t/(1-t))$, this implies

$$t(1-t)J_m' + (m+1+mt)J_m = 1.$$

Using (18) gives (16). The derivations may first be made for $0 < t < 1$; all resulting identities are polynomial and consequently hold at the endpoints as well. □

For explicit computation one can use the equivalent finite expansion

$$Y_n(t) = \sum_{j=0}^{n-2} \frac{(-1)^j (j+1)(n-2)!}{(n-2-j)!(n+j)!} t^j. \tag{19}$$

It follows directly by expanding the integrand in (13) and using the beta integral.

6 Reduction to a single moment

Let

$$M_n = \int_0^1 w_n Y_n \, \mathrm{d}t, \qquad B_n = \int_0^1 w_n \, \mathrm{d}t = \frac{3}{2n(n+1)(2n+1)(2n+3)}.$$

Lemma 4. *With*

$$u_1 = -\frac{8(n-1)(2n-1)(3n+1)}{(4n-1)(4n+1)}, \qquad u_0 = \frac{(n+1)(2n+3)(6n-1)}{n(4n-1)(4n+1)},$$

we have

$$a_n = \frac{K_n}{2n-1} \left(C u_1 M_n + C^2 u_0 B_n \right). \tag{20}$$

Proof. The explicit rational functions h, g, k in Appendix A satisfy

$$w_n \mathcal{H}_n = \frac{\mathrm{d}}{\mathrm{d}t} \{ w_n (h Y_n^2 + C g Y_n + C^2 k) \} + w_n (C u_1 Y_n + C^2 u_0). \tag{21}$$

This is a rational algebraic identity after using (16); the appendix gives the coefficient identities that verify it. Each of h, g, k is bounded on $[0,1]$, and Y_n is a polynomial. Since w_n vanishes at both endpoints, the derivative integrates to zero. Equations (14) and (21) prove the result. □

7 A recurrence for the moment

Define

$$r_n = \frac{n(2n+1)(4n+5)}{3(2n-1)(3n+2)(3n+4)(4n+1)},$$
$$f_n = \frac{(2n+1)(28n^2+26n+1)}{3(n+1)(2n-1)(3n+2)(3n+4)(4n+1)}. \tag{22}$$

Lemma 5. *For every $n \geq 2$,*

$$M_{n+1} + r_n M_n = C B_n f_n. \tag{23}$$

Proof. Put

$$\alpha = \frac{2n(2n+1)(4n+5)}{3(3n+2)(3n+4)(4n+1)}$$

and $S(t) = (s_0 + s_1 t + s_2 t^2 + s_3 t^3)/(1+t)^2$, where

$$s_0 = -\frac{2(2n+1)(4n+5)}{3(3n+2)(3n+4)(4n+1)}, \qquad s_1 = \frac{11n^2+26n+14}{3(n+1)(3n+2)(3n+4)},$$
$$s_2 = \frac{2n+1}{3(n+1)(3n+4)}, \qquad s_3 = -\frac{1}{3n+4}.$$

Direct differentiation and collection of powers of t verify

$$P S' + [n - (2n-1)t - 3(n+1)t^2] S = t(1-t)^2 - \alpha. \tag{24}$$

For example this verification requires only multiplication by $(1+t)^3$ and the displayed denominators, followed by equality of polynomial coefficients. With $\ell = w_n'/w_n = (2n-1)/t - 3/(1-t)$, the bracket is $P' + \ell P - R$. Thus (16) gives

$$\frac{\mathrm{d}}{\mathrm{d}t}(w_n P S Y_n) = w_n\{[t(1-t)^2 - \alpha]Y_n + CS\}.$$

The boundary terms vanish, so

$$\int_0^1 t^{2n}(1-t)^5 Y_n \,\mathrm{d}t = \alpha M_n - C\int_0^1 w_n S \,\mathrm{d}t. \tag{25}$$

Here the rational integral also has an explicit elementary certificate. Set

$$D = 3(n+1)(3n+2)(3n+4)(4n+1),$$
$$H(t) = \frac{2n(1-t)-4t}{1+t} - \frac{t(1-t)}{(1+t)^2},$$
$$\gamma = -\frac{28n^2+33n+8}{D}, \qquad d_0 = \frac{2(20n^3+11n^2-11n-5)}{D}, \qquad d_1 = -\frac{1}{3n+4}.$$

Substitution gives $S = d_0 + d_1 t + \gamma H$, and

$$w_n H = \frac{\mathrm{d}}{\mathrm{d}t}\frac{t^{2n}(1-t)^4}{1+t}.$$

The latter function vanishes at both endpoints. Since $\mathrm{B}(2n+1,4) = nB_n/(n+2)$, it follows that

$$\int_0^1 w_n S \,\mathrm{d}t = \beta B_n, \qquad \beta = \frac{4n^4+33n^3-17n^2-60n-20}{3(n+1)(n+2)(3n+2)(3n+4)(4n+1)}. \tag{26}$$

Finally multiply (17) by $t^{2n}(1-t)^3$ and integrate. Its first term is $2(2n-1)M_{n+1}$, and its right side is $CB_n/(n+2)$. Substituting (25)–(26) gives

$$M_{n+1} + \frac{\alpha}{2(2n-1)}M_n = \frac{CB_n}{2(2n-1)}\left(\frac{1}{n+2} + \beta\right).$$

The two coefficients simplify to r_n and f_n in (22), respectively. This proves (23). □

8 Recovery of the proposed recurrence

Let $T_n = n!M_n$. Equations (2) and (20) give

$$c_n = U_n T_n + V_n, \tag{27}$$

where

$$U_n = -\frac{2n(3n+1)}{(2n-1)(4n+1)}, \qquad V_n = \frac{3(6n-1)}{8(n-1)(2n-1)^2(2n+1)(4n+1)}.$$

Equation (23) becomes $T_{n+1} + d_n T_n = e_n$, with

$$d_n = \frac{n(n+1)(2n+1)(4n+5)}{3(2n-1)(3n+2)(3n+4)(4n+1)},$$
$$e_n = \frac{28n^2+26n+1}{2(n+1)(2n-1)(2n+3)(3n+2)(3n+4)(4n+1)}.$$

For $A_n = 3(3n+1)(3n+2)$, $E_n = (n+1)^2$, and

$$G_n = \frac{3(7n^2-1)}{4n(n-1)(2n-1)^2(2n+1)^2},$$

the elementary rational identities

$$-A_nU_{n+1}d_n + E_nU_n = 0,$$
$$A_n(U_{n+1}e_n + V_{n+1}) + E_nV_n = G_n$$

now prove (3).

For completeness, the elimination to the exact target coefficients can be made entirely explicit. We have

$$\rho_n := \frac{G_{n+1}}{G_n} = \frac{(n-1)(2n-1)^2(7n^2+14n+6)}{(n+1)(2n+3)^2(7n^2-1)}.$$

Subtract ρ_n times (3) from its shift by one to get

$$A_{n+1}c_{n+2} + (E_{n+1} - \rho_n A_n)c_{n+1} - \rho_n E_n c_n = 0. \tag{28}$$

If $c_n = q_n a_n$ as in (2), then

$$\frac{q_{n+1}}{q_n} = \frac{(n-1)(n+1)^3}{8n(4n-1)(4n+1)(2n+1)} =: s_n.$$

After division by q_n, the three coefficients in (28) are $A_{n+1}s_ns_{n+1}$, $(E_{n+1}-\rho_nA_n)s_n$, and $-\rho_nE_n$. Multiplication by

$$\frac{64(2n+1)(2n+3)^2(4n-1)(4n+1)(4n+3)(4n+5)(7n^2-1)}{(n-1)(n+1)}$$

gives exactly the three coefficients in (1). This proves that equation for $n \geq 2$. The case $n = 1$ follows by direct substitution of $(1, 1, 8)$. All normalizing factors used for $n \geq 2$ are nonzero. The leading coefficient in (1) is positive for every integer $n \geq 1$, so together with a_1, a_2 the recurrence uniquely determines the whole combinatorial sequence. This completes the proof of Theorem 1.

9 Verification, reproducibility, and scope

The proof consists of finite sums, polynomial identities, and integrals of polynomials or bounded rational functions. The symbolic checks clear denominators and test zero polynomials over the rationals; they apply to symbolic n, not just sampled integer values. The original combinatorial count enters through the reversible cut in Section 3. No recurrence-generated values are used in that argument.

The accompanying implementation provides three separate checks: direct enumeration using cell predecessors in all four directions; the positive double sum; and an independent interlacing-integral/Pfaffian formula. The positive sum was evaluated for $1 \leq n \leq 100$, matched the retrieved OEIS table, and gave zero target residual through $n = 98$. Profile counting was checked through $n = 30$, and the separate cell-bitmask calculation through $n = 14$. The first values are

$$1,\ 1,\ 8,\ 169,\ 6392,\ 352184,\ 25097600,\ 2152061145.$$

These finite checks detect implementation mistakes; the proof of the infinite statement is the argument above. The source package contains commands, dependency versions, exact certificates, and file hashes. It can be verified without network access after installing its stated dependency.

This investigation was produced by GPT-6 Astra, an AI system. Its work included derivation, symbolic calculation, independent internal review, and manuscript preparation. No external peer review or proof-assistant formalization is claimed. The literature audit is bounded; this note establishes the mathematical result within the supplied proof, not a claim of first publication.

A The quadratic reduction certificate

The rational functions in (21) are

$$\begin{aligned} h &= \frac{2t(t-1)^2}{(4n-1)(t+1)^2}, \\ g &= \frac{t}{n(4n-1)(4n+1)(t+1)^2}\big[(16n^3-24n^2+8n)t^2 \\ &\qquad + (32n^3-32n^2+2n+3)t + 16n^3-40n^2-14n+3\big], \\ k &= -\frac{t}{2n(4n-1)(4n+1)(t+1)^2}\big[(12n^2+10n-2)t^2 \\ &\qquad + (24n^2+2n-1)t - 4n^2-12n+1\big]. \end{aligned}$$

Here P, R, C are as in Section 5 and u_1, u_0 as in Lemma 4. A short coefficient-level verification avoids any unspecified integration algorithm. Put

$$\ell = \frac{2n-1}{t} - \frac{3}{1-t}, \quad a = n + 3(n-2)t, \quad b = t[2n+1+(2n-7)t], \quad d = 2t^2(1-t),$$

and

$$Q_2 = a - \frac{bR}{P} + \frac{dR^2}{P^2}, \quad Q_1 = \frac{b}{P} - \frac{2dR}{P^2}, \quad Q_0 = \frac{d}{P^2}.$$

Replacing Y' by $(C - RY)/P$ in $\mathcal{H}_n$ gives $Q_2Y^2 + CQ_1Y + C^2Q_0$. The displayed h, g, k obey

$$\begin{aligned} h' + (\ell - 2R/P)h &= Q_2, \\ g' + (\ell - R/P)g + u_1 &= Q_1 - 2h/P, \\ k' + \ell k + u_0 &= Q_0 - g/P. \end{aligned}$$

Each equality follows by differentiating its explicit rational expressions and multiplying out the denominators. These three coefficient equalities are exactly (21). Their possible displayed poles in P^{-1} occur only in the intermediate verification: the actual integrated derivative uses the bounded functions h, g, k and the polynomial Y_n. Thus no endpoint term or limit is omitted.

References

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